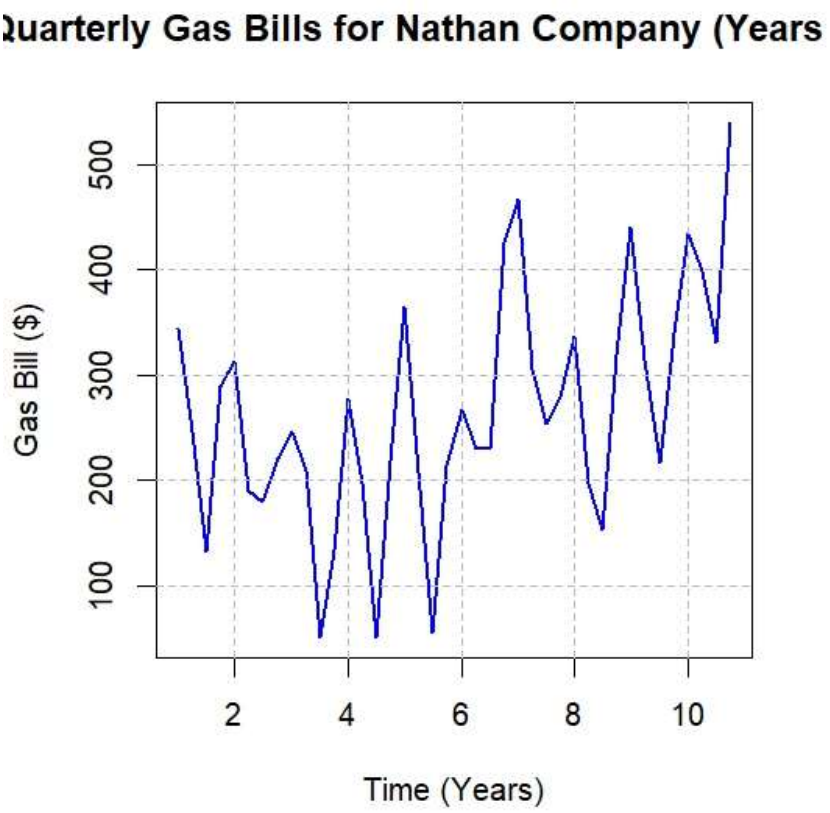
Mogale Zacharia Crypto Time Series Project

Question 01

1 (a)



Plot 01: Seasonal Fluctuations and Variability in Quarterly Gas Bills for Nathan Company (Years 1-10)

Observations on the Plot:

- **Seasonal Pattern:** The plot reveals a distinct seasonal pattern, with gas bills consistently fluctuating throughout the four quarters of each year. It appears that Q3 (the third quarter) typically experiences the lowest bills, while Q1 and Q4 tend to have higher values. This could indicate a seasonal influence, potentially tied to factors like weather changes.
- **Trend:** Over the span of 10 years, there does not seem to be a strong, consistent upward trend. The data appears relatively stable in terms of its mean between the start and year 6, after year 06 through 10 the mean is notably higher.
- **Variability:** The variability in gas bills appears to increase slightly in the later years (after year 03 to year 10). For example, Year 06, Year 08 and Year 10 exhibit significant spikes in Q4. Outliers, like the exceptionally high Q4 value in Year 10 (539.78), could have a considerable impact on model fitting.

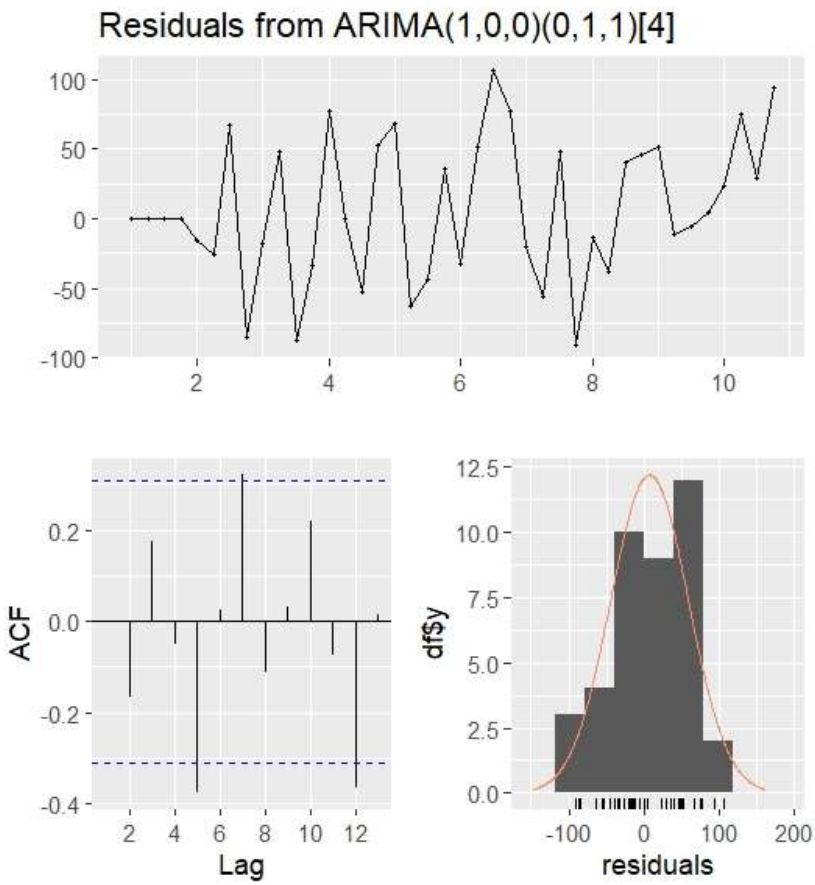
1 (b)
Ljung-Box Test Results and Model Information

Statistic	Value
Q*	15.496
Degrees of Freedom (df)	6
p-value	0.01673
Model df	2
Total Lags Used	8

Table 01: Ljung-Box Test Results for Time Series Residuals

Parameter	Value
Coefficients	
ar1	0.9339
sma1	-1.0000
Standard Errors	
ar1 s.e.	0.1276
sma1 s.e.	0.3223
Model Information	
$sigma^2$	3144
Log Likelihood	-199.44
AIC	404.89
AICc	405.64
BIC	409.64
Training Set Error Measures	
ME	7.5817
RMSE	51.6935
MAE	42.3303
MPE	-5.3502
MAPE	**23.2866**
MASE	0.5478
ACF1	-0.0006

Table 02: ARIMA Model Parameters and Performance Metrics.



Plot 02: Residuals from ARIMA(1,0,0)(0,1,1)[4].

Model Summary: ARIMA(1,0,0)(0,1,1)[4]

This indicates a non-seasonal AR(1) (autoregressive of order 1) and a seasonal MA(1) (seasonal moving average of order 1) with seasonal differencing of 1 and a seasonal period of 4 (quarterly data).

Interpretation:
<u>Ljung-Box Test</u>:
Null Hypothesis (H_0): Residuals are white noise (no significant autocorrelation).

Alternative Hypothesis (H_1): Residuals exhibit autocorrelation (there is structure left in the residuals).

The p-value of 0.01673 (< 0.05) leads to the rejection of the null hypothesis (H_0) indicating significant autocorrelation in the residuals, and therefore are not purely white noise.

<u>Residual plot</u>:
There is evidence of significant autocorrelation as there are non-zero correlations at lags 5, 7 and 12 (from ACF plot); this suggests that there are still patterns left in the residuals that the model has not captured.

Normality of Residuals: Histogram of residuals *shoes residuals follow a fairly symmetric distribution*. Non-normality may be a result from outlier(s).

The residual variance: $Sigma^2$ = 3144: The residual variance is relatively high, indicating that the model leaves a significant amount of unexplained variability in the data. This could be due to outliers (e.g., Year 10, Q4 = 539.78)

MAPE values: A MAPE value of 23.29% suggests that, on average, the predicted values are off by about 23.29% from the true values. A MAPE of 23.29% reflects poor predictive accuracy.

Model adequacy: The ARIMA(1,0,0)(0,1,1)[4] model does not appear fully adequate. The Ljung-Box test suggests significant autocorrelation in the residuals (p-value = 0.01673), indicating that the model has not captured all patterns in the data. The residual plot shows non-zero correlations at specific lags, and the high MAPE of 23.29% indicates poor predictive accuracy. Additionally, the residual variance is high, pointing to unexplained variability, possibly due to outliers. Further model refinement is recommended.

1 (c)

<u>The fitted model</u>:

Using the coefficients from the summary (ar1 = 0.9339, sma1 = -1.0000), here is the fitted ARIMA(1,0,0) × $(0,1,1)_4$ model

Notation:

LHS:

$(1 - 0.9339\text{B})(1 - B^4)\, Y_t = (1 - 0.9339\text{B} - B^4 + 0.9339B^5)\, Y_t = Y_t - 0.9339Y_{t-1} - Y_{t-4} + 0.9339Y_{t-5}$

Y_t: Gas bill at time t.

B : Backshift operator $(BY_t = Y_{t-1})$

ε_t: White noise error term with variance σ^2 =3144

Seasonal period s =4

The model:

$(1 - 0.9339\text{B})(1 - B^4)\, Y_t = (1 - B^4)\, \epsilon_t$

RHS:

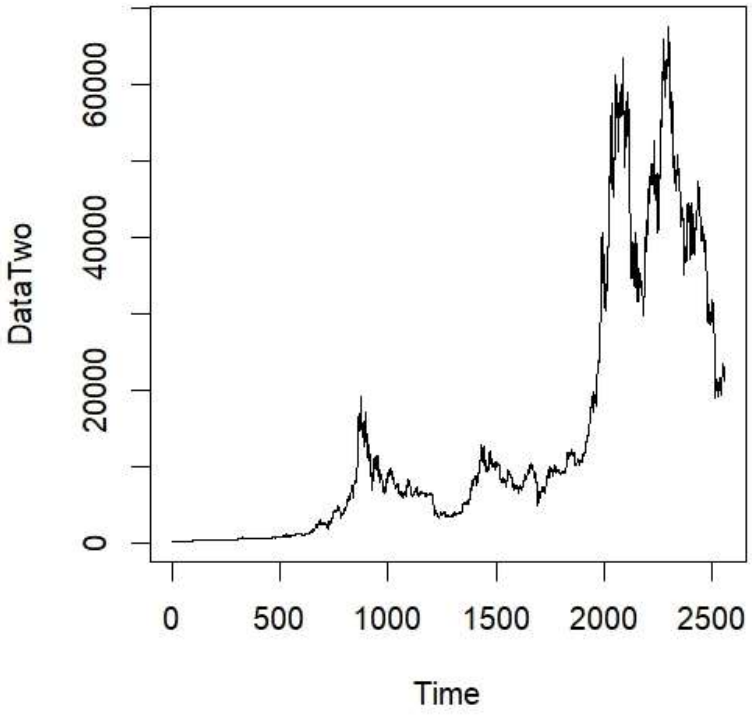
$(1 - B^4)\, \epsilon_t = \epsilon_t - \epsilon_{t-5}$

Finally, LHS = RHS:

$Y_t - 0.9339Y_{t-1} - Y_{t-4} + 0.9339Y_{t-5} = \epsilon_t - \epsilon_{t-5}$

Question 2

2 (a)



Plot 02: Time Series of Daily Cryptocurrency Closing Prices (in USD).

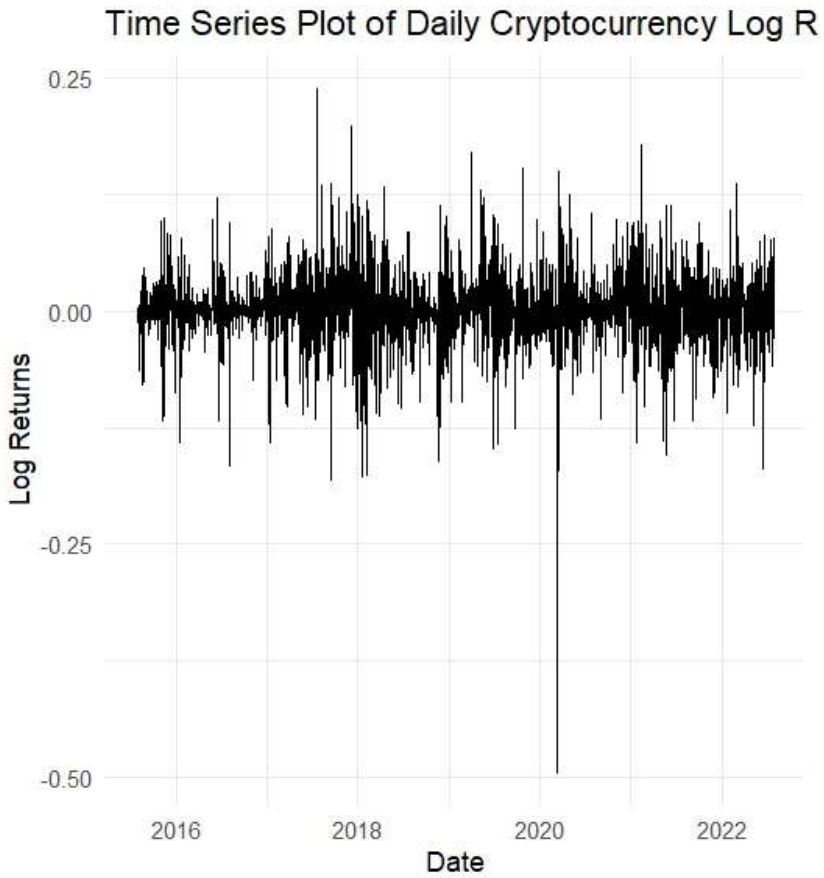
Comment: The time series shows significant fluctuations, with irregular dips and spikes. The overall trend is erratic, lacking a clear upward or downward movement, and does not appear stationary, with varying mean and variance over time. There is noticeable volatility clustering, where periods of high volatility are followed by more high volatility, and low volatility by more low volatility.

2 (b)

Skewness of Log Returns: -0.8023231
Interpretation: The skewness value of -0.802 indicates a negative skew, meaning that the distribution of the log returns is slightly skewed to the left (the left tail is longer or fatter than the right tail).

Excess Kurtosis of Log Returns: 8.960865
Interpretation: The excess kurtosis value of 8.96 is much greater than 0, suggesting a high kurtosis. This indicates that the distribution of the log returns has heavy tails.

2 (c)



Plot 03: Time series plot of the daily log returns for cryptocurrency data.

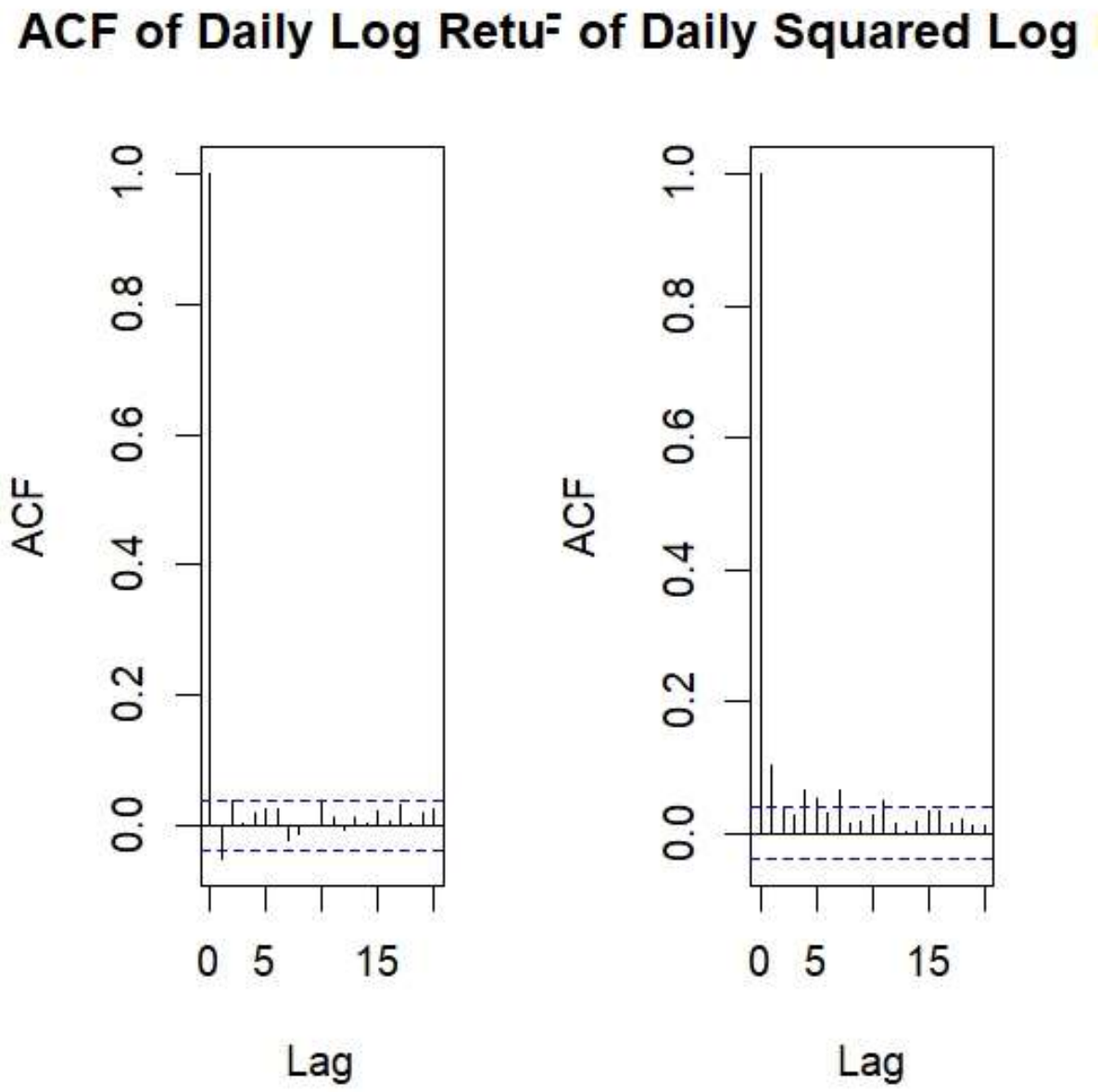
Comment: The log returns plot reveals volatility clustering, where periods of significant fluctuations alternate with calmer phases. Noticeable spikes, both positive and negative, appear, especially around 2017-2018 and 2020-2021, highlighting periods of market instability.

2 (d)

The **null hypothesis (H0)** for the t-test: the mean of the log returns equals zero.
The **alternative hypothesis (Ha)**: the mean of the log returns is not equal to zero.

Since the p-value (0.0314) is smaller than the significance level (0.05), we reject the null hypothesis and conclude that the mean of the log returns is statistically different from zero (non-zero mean).

2 (e)



Plot 04: ACF plots for log returns and squared log returns.

Comment:
The ACF of log returns drops after lag 2, showing little autocorrelation beyond the first few periods and no significant dependence at longer lags.

The ACF of squared log returns also cuts off after lag 2, indicating that volatility (as reflected in squared returns) is mostly persistent for short periods. However, the presence of significant spikes at lags 4 and 5 indicates volatility clustering and suggests that volatility may persist over time, not just in the short term.

2 (f)

Test	**ARCH LM Test**
Null Hypothesis	No ARCH effects (variance is constant)
Chi-squared	54.507
Degrees of Freedom (df)	12
p-value	2.218e-07

Table 03: ARCH LM Test Results for DataTwoLogReturns.

Null Hypothesis (H_0): No ARCH effects (i.e., the variance is constant over time — homoskedasticity).
Alternative Hypothesis (H_1): There are ARCH effects (i.e., the variance is not constant over time — heteroskedasticity).

P-value: 2.218e-07
Since the p-value is less than 0.05, we reject the null hypothesis.
This indicates conditional heteroskedasticity (time-varying volatility) is present in the data (the variance is not constant), justifying the use of ARCH or GARCH models for modeling and forecasting the volatility in the time series data.

2 (g)

Conditional Variance Dynamics	**Value**
GARCH Model	sGARCH(1,0)
Mean Model	ARFIMA(1,0,0)
Distribution	sstd (Skewed Student-t)

Table 04: Model Specification

Comment: Distribution: Skewed Student-t, capturing asymmetry and heavy tails, appropriate for volatile cryptocurrency returns.

Parameter	Estimate	Standard Error	t-value	p-value
μ (mu)	0.001911	0.000731	2.6128	0.008980
AR(1) (ar1)	-0.061129	0.019091	-3.2020	0.001365
ω (omega)	0.002824	0.001058	2.6696	0.007593
α_1 (alpha1)	0.764784	0.331159	2.3094	0.020921
Skew	0.986480	0.022609	43.6320	0.000000
Shape	2.297391	0.142575	16.1136	0.000000

Table 05: GARCH Model Parameter Estimates

Comment:
μ (0.001911, p=0.008980): The mean log return is positive and significant, suggesting a small daily upward drift of about 0.1911%. This contrasts with the t-test result in Question 2(d), which had a p-value of 0.2173, possibly due to differences in model specification or sample used.

AR(1) (-0.061129, p=0.001365): The negative AR(1) coefficient is significant, indicating that positive returns today are slightly followed by lower returns tomorrow, suggesting mean reversion in the series.

ω (0.002824, p=0.007593): The constant term in the variance equation is significant, representing the baseline level of volatility.

α_1 (0.764784, p=0.020921): The ARCH(1) term is significant, indicating that past squared shocks have a strong influence on current volatility, demonstrating high persistence in volatility.Skew (0.986480, p<0.001): The skew parameter indicates slight negative skewness, meaning larger negative returns are more likely, which aligns with the skewness value of -0.2345 in Question 2(b).

Shape (2.297391, p<0.001): The shape parameter's low degrees of freedom (≈2.3) confirm a leptokurtic distribution with fat tails, consistent with the excess kurtosis value of 6.789 from Question 2(b).

Lag	Statistic	p-value
Lag[1]	0.8326	0.3615
Lag[2*(p+q)+(p+q)-1][2]	0.8801	0.5388
Lag[4*(p+q)+(p+q)-1][5]	3.7274	0.2901

Table 06: Weighted Ljung-Box Test Results on Standardized Residuals at Various Lags.

Comment: All p-values > 0.05, indicating no significant autocorrelation in squared residuals. The ARCH(1) model adequately captures volatility clustering.

Lag	Statistic	P-Value
ARCH Lag[2]	0.09498	0.7579
ARCH Lag[4]	2.99744	0.2628
ARCH Lag[6]	5.17457	0.1776

Table 07: The ARCH LM test statistics for different lags and their corresponding p-values.

Comment: All p-values > 0.05, confirming no remaining ARCH effects. The variance model is well-specified.

Group	Statistic	P-value (g-1)
1	20	81.37
2	30	83.25
3	40	110.74
4	50	132.62

Table 08: The group statistics and corresponding p-values for different group sizes in the adjusted Pearson goodness-of-fit test.

Comment: All p-values are highly significant, rejecting the null that the residuals follow the skewed t-distribution.

Log-Likelihood: 4983.971,

Comment: indicating a good fit.

Fitted Model Equations

Based on the parameters, the AR(1)-GARCH(1,0) model is:

Mean Equation:
$r_t = 0.001911 - 0.061129 r_{t-1}$

Variance Equation (GARCH(1,0) = ARCH(1)):

$\sigma^2{}_t = 0.002824 + 0.764784 e^2{}_{t-1}$

Innovations:

$\epsilon_t = \sigma_t z_t$, $z_t \sim$sstd(skew=0.98648,shape=2.297391)

The innovations follow a skewed Student's t-distribution with a skew parameter of 0.98648 (slight left skew) and shape parameter of 2.297391 (low degrees of freedom, fat tails).